Exploring the Functional Diversity of the L7-like Ribosomal Protein

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Introduction

- Ribosomal proteins interact with ribosomal RNAs to increase ribosome stability
- Previous work shows ribosomal proteins may exhibit **ribosomal exaptation**, where ribosomal components occur in extra-ribosomal systems & play a unique, different role than the original function [1].
- Some ribosomal proteins stabilize the guide RNA of RNA-guided systems like the CRISPR-Cas12k, C/D Box, & H/ACA Box systems.
- **L7** is a ribosomal protein in the large ribosomal subunit.
- L7 or paralogs of L7 interact with:
  - **C/D Box**, which is an RNA-guided methyltransferase
  - **H/ACA Box**, which is an RNA-guided pseudouridylation
- Paralogs of L7 can binds guide RNAs in some species
  - **Homologs/Paralogs:** similar genes that share an origin.

Research Question: Can we use L7 domain as a seed to discover novel biological systems featuring RNA components?
Research Goal: Apply structural & sequence mining to explore functional diversity of L7-like domain

Methods

1) Structural comparison of L7 in different systems like the C/D Box, H/ACA Box, & bacterial ribosome.
2) Mine L7 protein against Alpha Fold databases (Uniprot, MGnify, & BFVD) with FoldSeek, HHPred, & DALI [8].
3) Phylogenetic analysis to explore diversity of L7 homologs.
4) Explore relevant family using domain architectures & embedding visualization analysis.

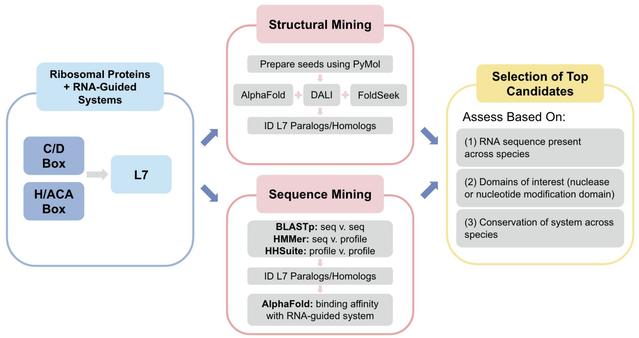
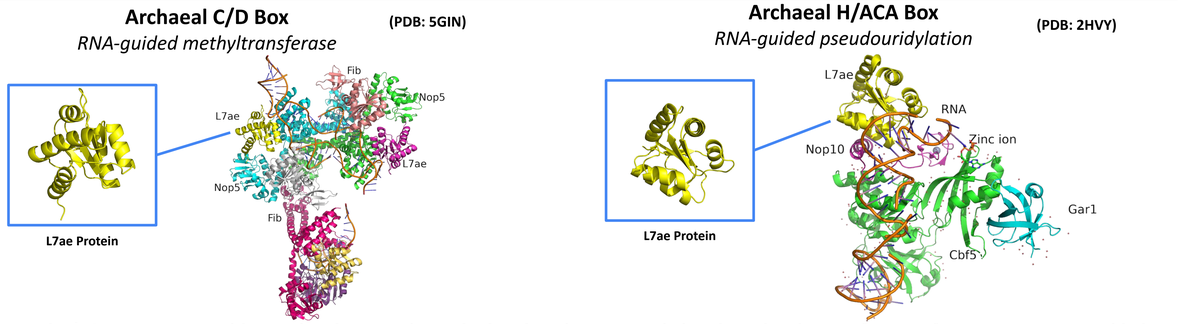


Figure 1: Methodology pipeline, starting with L7 protein input into structural and sequence mining pipelines. Top candidates are assessed based on validity and requirements for use in biological applications.

Results

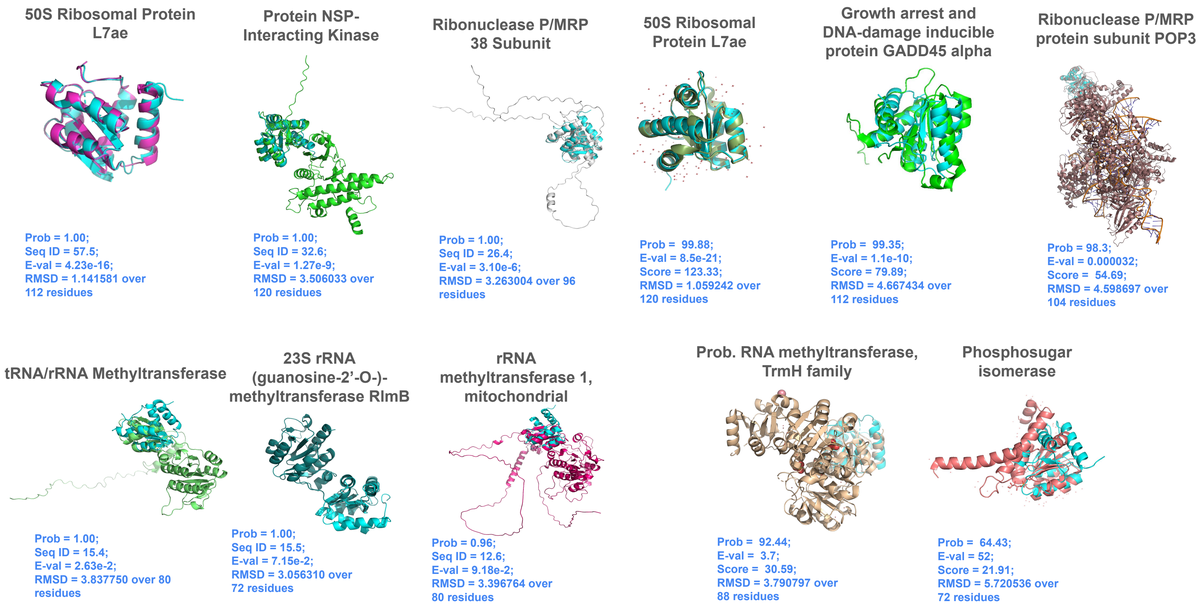
Structural Comparison



C/D Box and H/ACA Box systems with associated L7 proteins and other domains. L7 protein is enlarged and PBD references for structures are provided.

Structural Mining (Foldseek) | Sequence Mining (HHSuite)

Examples of candidates identified through mining. L7 protein is aligned against each candidate in blue.



Probability: confidence level of prediction. **Sequence Identity:** percent of identical amino acids. **E-value:** number of matches we would expect to find by chance. **RMSD**: average distance between atoms of superimposed structures.

Conclusion

- Structural comparison shows L7 can interact with various RNA and protein partners
- A large diversity of proteins encode L7-like domain
- Domain architecture analysis reveals functions of interest e.g. methyltransferase, ribonuclease, etc.

Future Work

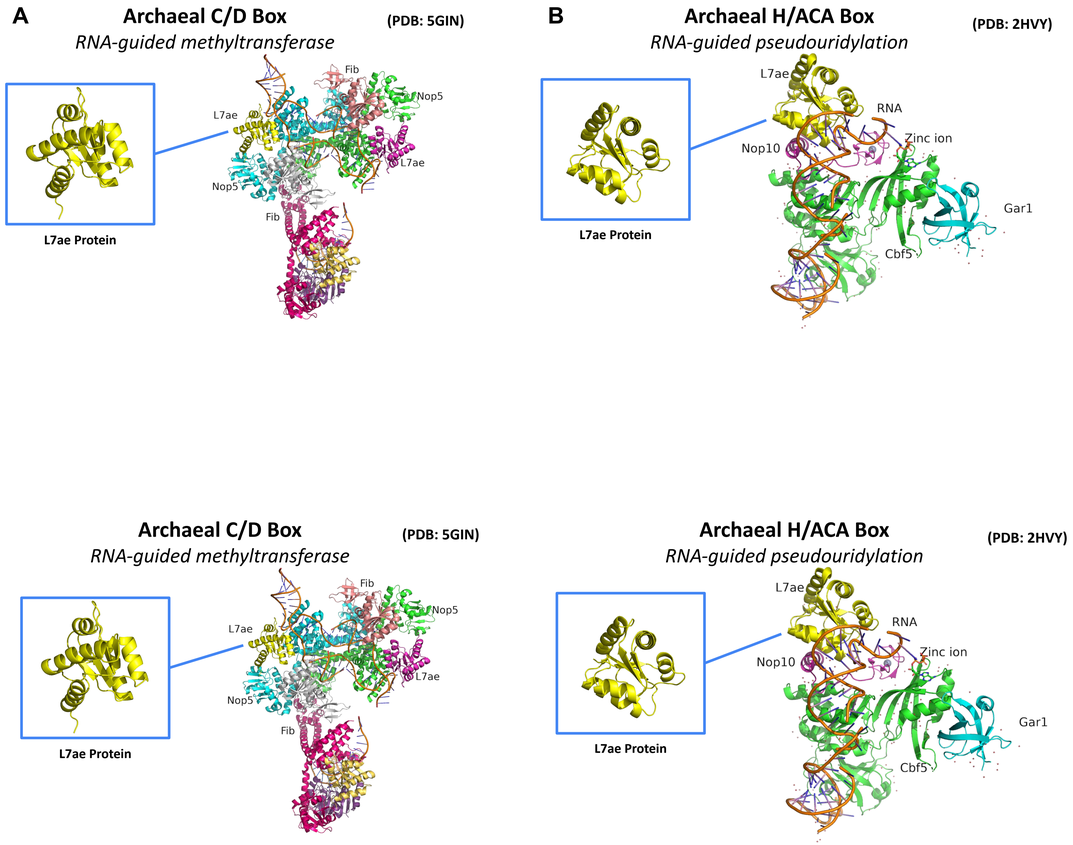
- Run large scale structural mining using FoldSeek coupled to DALI on Uniprot, Magnify, BFVD.
- Phylogenetic analysis & embedding projection analysis to identify families based on evolutionary ancestry.
- Domain architecture analysis to explore functions of different components of the protein.

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References

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A
Archaeal C/D Box
RNA-guided methyltransferase
(PDB: 5GIN)
L7ae Protein
Fib
Nop5
L7ae
L7ae
Nop5
Fib
B
Archaeal H/ACA Box
RNA-guided pseudouridylation
(PDB: 2HVY)
L7ae Protein
L7ae
RNA
Zinc ion
Nop10
Gar1
Cbf5
Archaeal C/D Box
RNA-guided methyltransferase
(PDB: 5GIN)
L7ae Protein
Fib
Nop5
L7ae
L7ae
Nop5
Fib
Archaeal H/ACA Box
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